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Definitions

BFS (Builders FirstSource) = BLDR.com, NYSE BLDR, the building-products / component / distribution company.

Introduction / Purpose

Builders FirstSource (BFS) distributes a huge range of building-products for single-family, multi-family and light commercial construction. A clear understanding of core product categories, typical dimensions, strengths and naming conventions allows designers, estimators and field crews to make better choices. This guide organizes the major product families – dimensional lumber, engineered wood products (EWP), roof trusses, wall panels, siding, roofing and millwork – and explains when and why each type is used. It concludes with cross-category comparisons, decision guidelines and common questions.

Major Product Categories

Dimensional lumber

Dimensional lumber refers to solid-sawn boards cut to standard nominal sizes and surfaced four sides (S4S). After planing and drying, the boards are smaller than their nominal designations. Tables from industry associations show the actual dressed dimensions. For example, a nominal 2 × 4 board finishes about 1½ × 3½ inches, while a nominal 2 × 6 finishes 1½ × 5½ inches ¹. A table summarizing common sizes is provided below.

Typical dimensions

Nominal thickness × width	Actual size (approx.)	Use cases
1 × 4	~¾ × 3½ in ²	Boards, shelving
2 × 4	~1½ × 3½ in ¹	Wall studs, plates
2 × 6	~1½ × 5½ in ¹	Exterior walls, floor joists
2 × 8	~1½ × 7¼ in ²	Floor joists, rafters
2 × 10	~1½ × 9¼ in ²	Floor joists, beams

Nominal thickness × width	Actual size (approx.)	Use cases
2 × 12	~1½ × 11¼ in ²	Beams, stair stringers

Lengths and span capability

Lumber is stocked in a range of lengths. A supplier catalogue shows “stud-length” boards of 92½ in, 104½ in and 116½ in, plus standard 4-ft, 8-ft, 10-ft, 12-ft, 14-ft, 16-ft, 18-ft, 20-ft, 22-ft, 24-ft, 26-ft, 28-ft and 32-ft lengths for 2 × 4 through 2 × 12 boards ³. Spans depend on species and load; a quick reference shows that a 2 × 6 floor joist spaced 12 inches apart can span about 11½ ft under a 30 psf live load, while a 2 × 10 can span almost 19¾ ft under a 40 psf live load ⁴. When spanning long distances, increasing thickness/height or switching to engineered wood may be necessary.

Strengths and limitations

- **Species and grade:** Douglas fir–larch, southern yellow pine (SYP) and SPF (spruce–pine–fir) are common species. Higher grades have fewer knots and provide higher allowable bending stresses.
- **Moisture content (MC):** Kiln-dried (KD) lumber is dried to about 19 % moisture (marked “KD” or “KD 19”). Higher moisture reduces strength and may cause shrinkage, so avoid using green lumber in finish applications.
- **Span and deflection:** Depth controls stiffness; deeper boards deflect less for the same load. When spans exceed design limits, switch to EWP or trusses.

Engineered wood products (EWP)

EWP includes I-joists and structural composite lumber (SCL) such as laminated veneer lumber (LVL), laminated strand lumber (LSL) and parallel-strand lumber (PSL). These products use veneers or strands bonded with adhesives, creating consistent and high-strength members.

I-joists

The APA notes that I-joists are available in depths such as 9½ in, 11¾ in, 14 in and 16 in with flange widths ranging from 1½ in to 3½ in ⁵. They are primarily used for floor and roof framing and have published span tables (e.g., APA PRI-400) that allow long spans with minimal deflection ⁶. The thin OSB or plywood web allows holes for ducts or pipes. Because I-joists are engineered, they are lighter and straighter than solid lumber.

Structural composite lumber (SCL)

SCL combines veneers or strands oriented in parallel or random patterns. The APA lists thicknesses from ¾ in to 3½ in for SCL products and notes that multiple pieces can be joined to make 3½ in or 5¼ in members that fit into 2 × 4 or 2 × 6 walls ⁷. LVL beams are widely used for headers and beams; one manufacturer’s catalog states that LVL headers/beam widths are 1¾ in, 3½ in, 5¼ in and 7 in with depths from 9½ in to 24 in ⁸. LVL framing members are available in widths 1½ in, 1¾ in and 2½ in and depths from 3½ in to 11¼ in ⁸.

When to choose EWP over solid sawn:

1. **Long spans or heavy loads:** I-joists and LVL beams can span greater distances than sawn lumber; for example, a deep LVL beam may carry loads that would require a 2 × 12 or built-up lumber beam.
2. **Consistency and stability:** Engineered products are manufactured to tight tolerances, reducing warp and twist and producing a flatter floor or roof.
3. **Availability of long lengths:** I-joists and LVL can be produced in lengths up to 60 ft or more, reducing splicing and joint labour.
4. **Environmental efficiency:** Using veneers/strands from smaller trees reduces waste compared with large dimension lumber.

Roof trusses

Roof trusses are pre-fabricated triangular assemblies that replace stick-framing rafters. A construction article notes that trusses are typically spaced between 16 in and 24 in apart for residential roofs, with 24 in on-center common for gable roofs; heavy snow or wind may warrant 16 in spacing, while some commercial buildings may allow spacing up to 48 in ⁹. Trusses use metal connector plates to join top chords, bottom chords and web members. Common types include fink (for gable roofs), hip, scissor (vaulted ceilings) and attic trusses. Advantages include speed of erection, long clear spans and design flexibility. Limitations include transportation restrictions (width and depth) and the need for engineered design.

Wall panels / sheathing

Wall panels often refer to factory-built wall sections or the sheathing products applied to them. Oriented Strand Board (OSB) is the dominant sheathing. OSB is made from cross-oriented wood strands bonded with heat-cured adhesives, creating a panel with high strength and stability; the APA states that OSB resists deflection, delamination, warping and racking ¹⁰. The performance categories include thicknesses of 3/8 in, 7/16 in, 15/32 in, 19/32 in, 23/32 in, 7/8 in and 1½ in, and common panel sizes are 4 × 8 ft, 4 × 9 ft and 4 × 10 ft; OSB is also manufactured in 8 × 24 ft panels or larger and can be custom-cut ¹⁰. Panels are used for wall and roof sheathing, subflooring and webs for I-joists.

Siding

Siding protects the building envelope and provides aesthetics. Fiber-cement and engineered wood siding are common. A manufacturer describes lap siding widths: 6-inch lap siding is suited to small or single-story homes or as an accent; 8-inch lap siding is the most commonly used size; and 12-inch lap siding is used for larger structures with tall walls ¹¹. Lap siding boards overlap to shed water; narrower boards create more visual texture while wider boards cover wall area more quickly.

Roofing materials

Asphalt shingles dominate residential roofing. A roofing guide explains that standard three-tab shingles measure approximately 12 × 36 inches, while architectural or dimensional shingles are larger at about 14 × 39 inches ¹². Three-tab shingles consist of three 12-in tabs separated by cutouts; they are lightweight and flat. Architectural shingles have multiple layers that create a thicker, more dimensional appearance and improved durability ¹². Specialty shingles include impact-resistant or solar-reflective varieties.

Millwork

Millwork covers doors, mouldings, windows and other interior trim. For interior doors, a home improvement guide states that pre-hung doors typically come 32 in or 36 in wide with thickness around 1¾ in ¹³. Slab doors (just the panel) are sold in widths of 24 in, 28 in, 30 in, 32 in and 36 in ¹³. Bedroom and bathroom doors are usually 30–32 in wide and 80 in tall, while closet doors range from 24 in to 36 in wide; smaller closets may use 18–24 in widths ¹³. French door pairs range from 48 in to 72 in wide ¹³. The rough opening should be 2 in wider and 2½ in taller than the door to fit the jamb ¹³. Other millwork includes baseboards, casings and crown mouldings, typically sized 2–8 in wide and made from MDF, pine or PVC.

Relationships and overlaps

Selecting between product categories involves understanding performance, cost and constructability. The following guidelines use chain-of-thought reasoning to help choose appropriate products:

1. **Identify loads and spans:** Start by calculating loads (dead + live) and required spans. If the span can be achieved with dimensional lumber within deflection limits and local codes, use solid sawn lumber for simplicity and lower cost. For longer spans or heavy concentrated loads, consider LVL beams or I-joists.
2. **Consider moisture and environmental conditions:** In wet areas or where shrinkage could cause finish problems, choose kiln-dried lumber or engineered wood. OSB and EWP perform consistently but need protection from moisture until enclosed.
3. **Evaluate installation complexity:** Prefabricated trusses and wall panels reduce site labour and speed construction. However, they require advance design and adequate crane access. Stick framing with dimensional lumber offers field flexibility but may be slower.
4. **Assess aesthetics and durability:** For exterior cladding, decide whether narrow (6 in) lap siding for detailed shadow lines or wider (8 in or 12 in) boards for fewer seams ¹¹. For roofing, choose between economical three-tab shingles or thicker architectural shingles for improved appearance and wind resistance ¹².
5. **Review code requirements and compatibility:** Ensure that EWP is installed per manufacturer span tables and that truss designs are engineered. Use appropriate fasteners and connectors (e.g., hangers for LVL beams, truss plates).

Naming conventions and suffixes

Construction products often use abbreviations and suffixes. Knowing these helps interpret plans and catalogues:

- **KD:** Kiln-dried lumber, dried to about 19 % moisture (e.g., “KD19”).
- **MC:** Moisture content; “MC 15” indicates lumber dried to 15 %.
- **S4S:** Surfaced four sides; the board has been planed on all faces.
- **PT:** Pressure-treated; lumber impregnated with preservatives for decay resistance.
- **LVL / LSL / PSL:** Laminated Veneer Lumber, Laminated Strand Lumber and Parallel-Strand Lumber – types of SCL.
- **OSB:** Oriented Strand Board.

- **HDO / MDO:** High-density/medium-density overlay plywood; plywood with resin-impregnated faces for smooth finish.

Tips and caveats

- **Species variation:** Southern yellow pine is strong but may shrink more; Douglas fir offers good strength and stability; SPF is lighter but has lower bending strength. Always check grade stamps and design values.
- **Moisture control:** Store lumber and panels under cover and off the ground. OSB edges may swell if exposed; use tongue-and-groove edges for subflooring and leave expansion gaps.
- **Allowable spans:** Resist the temptation to overspan dimensional lumber. Consult span tables (e.g., 2 × 10 SPF spaced 16 in o.c. may span ~15 ft for 40 psf live load ⁴). For floors requiring less bounce, consider using I-joists or LVL beams.
- **Fire and code requirements:** Some jurisdictions require fire retardant treatment or rated assemblies; check local codes. Engineered wood may require fire-rated sheathing or gypsum board coverings.

SCAMPER exploration – brainstorming future innovations

To inspire product innovation, apply the SCAMPER framework:

- **Substitute:** Could fiber-reinforced polymers replace wood web material in I-joists to increase moisture resistance? Could bamboo fibers substitute for wood strands in OSB to improve sustainability?
- **Combine:** Imagine combining engineered trusses with integrated wiring chases or ductwork to reduce trades conflicts. Combine siding panels with insulation to create one-step cladding.
- **Adapt:** Adapt LVL technology to produce curved beams or arches for architectural accents. Adapt OSB manufacturing to create pre-finished interior panels.
- **Modify:** Modify lap siding profiles by adding interlocking edges to improve weather-tightness. Modify roofing shingles with integrated solar cells.
- **Put to other use:** Use short offcuts of LVL or LSL as blocking or stair stringers. Repurpose scrap truss plates for garden structures.
- **Eliminate:** Eliminate traditional truss webs by using open-web steel chords combined with wood chords for hybrid systems. Eliminate field cutting by using fully prefabricated wall panel systems.
- **Reverse:** Reverse the typical assembly order by installing roof trusses before exterior wall panels to speed “dry-in.” Reverse the orientation of lap siding to create a modern reverse board-and-batten look.

Related topics / crosslinks

- **Fasteners and connectors:** proper nails, screws, hangers and straps for various products.
- **Insulated wall panels:** structural insulated panels (SIPs) that combine OSB and foam insulation.
- **Green building certifications:** how FSC-certified lumber and low-emission panels support sustainability goals.
- **Estimating software:** digital tools for take-offs and ordering of lumber, EWP and trusses.

Likely user questions

1. **What standard sizes of LVL beams are available?** LVL headers and beams come in widths of 1¾ in, 3½ in, 5¼ in and 7 in and depths ranging from 9½ in to 24 in ⁸.
2. **When should I choose an I-joist instead of a 2 × 12?** Use I-joists when spans or loads exceed the capacity of solid lumber; common I-joist depths (9½ in, 11¾ in, 14 in, 16 in) provide longer spans and consistent performance ¹⁴.
3. **What does “KD 19” mean on a lumber stamp?** It means the lumber is kiln-dried to approximately 19 % moisture content; kiln drying improves stability and reduces shrinkage.
4. **How far apart should roof trusses be spaced?** Residential roof trusses are typically spaced 16 in or 24 in on center; heavy snow/wind loads may require 16 in spacing, while some commercial roofs may permit spacing up to 48 in ⁹.
5. **What sizes do OSB panels come in?** OSB sheathing is available in thicknesses from 3/8 in to 1½ in and panel sizes such as 4 × 8 ft, 4 × 9 ft and 4 × 10 ft; larger 8 × 24 ft panels are also manufactured ¹⁰.
6. **What is the difference between three-tab and architectural shingles?** Three-tab shingles are about 12 × 36 inches and have three separate tabs, making them flat and economical; architectural shingles are larger (~14 × 39 inches) and have multiple layers for a thicker, dimensional appearance ¹².
7. **How wide are standard interior doors?** Pre-hung interior doors commonly come in widths of 32 in or 36 in, while slab doors range from 24 in to 36 in ¹³.
8. **What does “S4S” mean when describing lumber?** S4S stands for “surfaced four sides,” indicating the board has been planed on all faces.
9. **How do I determine allowable floor joist spans?** Consult span tables: a 2 × 6 floor joist spaced 12 in o.c. can span around 11½ ft under a 30 psf live load, whereas a 2 × 10 can span nearly 19¾ ft under a 40 psf load ⁴.
10. **What are typical lap siding widths, and when should I use each?** Common lap siding widths are 6 in (for accents or small homes), 8 in (most common for general use) and 12 in (for large structures or tall walls) ¹¹.

This knowledge article compiles typical dimensions, strengths, naming conventions and decision guidelines for major building-products categories, equipping construction professionals to select the right product for every application.

¹ Standard Sizing - Southern Forest Products Association

<https://www.southernpine.com/resources/specifying-southern-pine-lumber/standard-sizing/>

² Lumber Dimensions and Size Measurements

<https://www.advantagelumber.com/sizes.htm>

³ product listing lumber length rev. 11-11-15.xls

<https://www.clc-tn.com/wp-content/uploads/product-listing-lumber-by-length.pdf>

⁴ Lumber Spans Quick Reference Card (U.S.)

<https://www.weyerhaeuser.com/application/files/3215/4223/2335/LB-4026.pdf>

⁵ ⁶ ¹⁴ I-Joist - APA – The Engineered Wood Association

<https://www.apawood.org/i-joist>

7 Structural Composite Lumber (SCL) - APA – The Engineered Wood Association

<https://www.apawood.org/structural-composite-lumber>

8 RedLam Laminated Veneer Lumber Specification Information

<https://www.redbuilt.com/products/laminated-veneer-lumber/details/>

9 Importance of Truss Spacing: Best Practices for Wood Roof Trusses

<https://rangertruss.com/blog/importance-of-truss-spacing-best-practices/>

10 Oriented Strand Board (OSB) - APA – The Engineered Wood Association

<https://www.apawood.org/osb>

11 Lap Siding - Diamond Kote® Siding System

<https://diamondkotesiding.com/siding/lap-siding/>

12 Understanding Shingle Size: A Comprehensive Guide to Asphalt Shingles | CoMo Premium Exteriors

<https://comoexteriors.com/understanding-shingle-size-comprehensive-guide-asphalt-shingles-central-mo/>

13 Interior Door Size Guide

<https://www.houseidea.com/blog/interior-door-sizes/>